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Assignment No: - 2 Subject: - Operating system Class:-B.Sc. (ECS)-II

Div :-(A,B)

Question 1 — FCFS, SJF AND SRTF

Consider the following 3 processes:

Process	Arrival time	Burst time
P1	0	8
P2	1	4
P3	2	6

Solve using:

- ✚ **Fcfs Scheduling**
- ✚ **Non-Preemptive Sjf**
- ✚ **Srtf (Preemptive Sjf)**

For each algorithm:

- ✚ **Draw the gantt chart.**
- ✚ **Calculate waiting time (wt) for each process.**
- ✚ **Calculate turnaround time (tat) for each process.**
- ✚ **Calculate average waiting time (awt).**
- ✚ **Calculate average turnaround time (atat).**

Question 2 — FCFS, SJF AND SRTF

Consider the following 4 processes:

Process	Arrival time	Burst time
P1	2	5
P2	0	4
P3	1	2
P4	4	1

Solve using:

✚ **Fcfs scheduling, Non-preemptive sjf, Srtf (preemptive sjf)**

Find:

✚ **Execution order, Gantt chart, Waiting time (wt), Turnaround time (tat)**

✚ **Average waiting time (awt)**

✚ **Average turnaround time (atat)**

Question 3 — FCFS, SJF AND SRTF

Consider the following processes:

Process	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	8

Solve using:

✚ **Fcfs**

✚ **Non-preemptive sjf**

✚ **Srtf**

Draw the gantt chart and calculate **wt, tat, awt and atat**.

Question 4 — FCFS, SJF AND SRTF

Five processes are given below:

Process	Arrival time	Burst time
P1	3	2
P2	4	4
P3	2	8
P4	1	6
P5	0	3

Find the following using **fcfs**, **sjf** and **srtf**:

- ✚ **Execution order**
- ✚ **Gantt chart**
- ✚ **Waiting time of each process**
- ✚ **Turnaround time of each process**
- ✚ **Average waiting time (awt)**
- ✚ **Average turnaround time (atat)**

Question 5 — FCFS, SJF AND SRTF

Consider the following 4 processes:

Process	Arrival time	Burst time
P1	0	6
P2	2	4
P3	4	5
P4	6	3

Solve using:

- ✚ **Fcfs**
- ✚ **Non-preemptive sjf**
- ✚ **Srtf**

Draw the gantt chart and calculate **wt**, **tat**, **awt** and **atat** for each scheduling algorithm.

Submission Deadline: Within 7 Days

1. All students are required to complete and submit the assignment **within 7 days** from the date of receiving this notice.
2. Please ensure that the assignment is complete, neat, and properly presented.